ANALYSIS REPORT

Feedforward Model Explanatory Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report documents the current `feedforward` TE baseline already implemented in the repository.

The goal is to frame the model quickly from three complementary angles:

- what the model is conceptually;
- how it behaves in the TE problem;
- how the Python implementation and training workflow are organized in this repository.

Model Description

The `feedforward` model is the simplest neural baseline currently available for TE regression.

It is a point-wise multilayer perceptron (MLP) that receives one TE sample at a time after curve-to-point conversion. Each point is described by:

- output angular position;
- input speed;
- input torque;
- oil temperature;
- direction flag.

The model maps these five inputs directly to a scalar TE prediction. This is the most generic static nonlinear regressor in the current program. It does not explicitly encode periodic structure, harmonic bias, or analytical decomposition.

Operating Principle

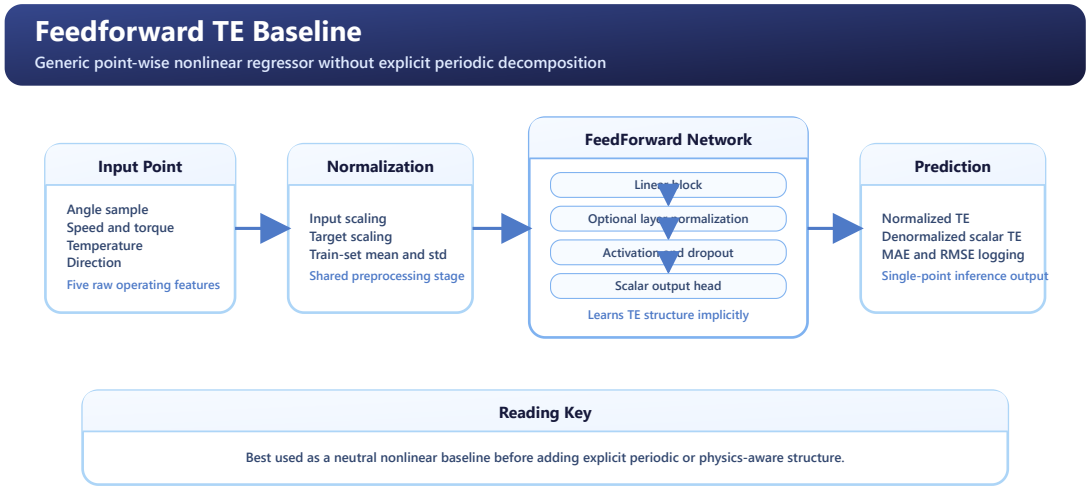
The operating principle is standard nonlinear regression in normalized feature space.

At training time

1. the datamodule flattens TE curves into point-wise samples;
2. input and target statistics are computed on the train split;
3. the MLP receives normalized input features;
4. the network outputs a normalized TE prediction;
5. loss is computed in normalized space, while `MAE` and `RMSE` are reported after denormalization.

This means the model learns a direct nonlinear mapping: `x -> f_theta(x) -> TE` without any explicit physics or periodic prior beyond what can be inferred from data.

Conceptual Map



Feedforward conceptual diagram

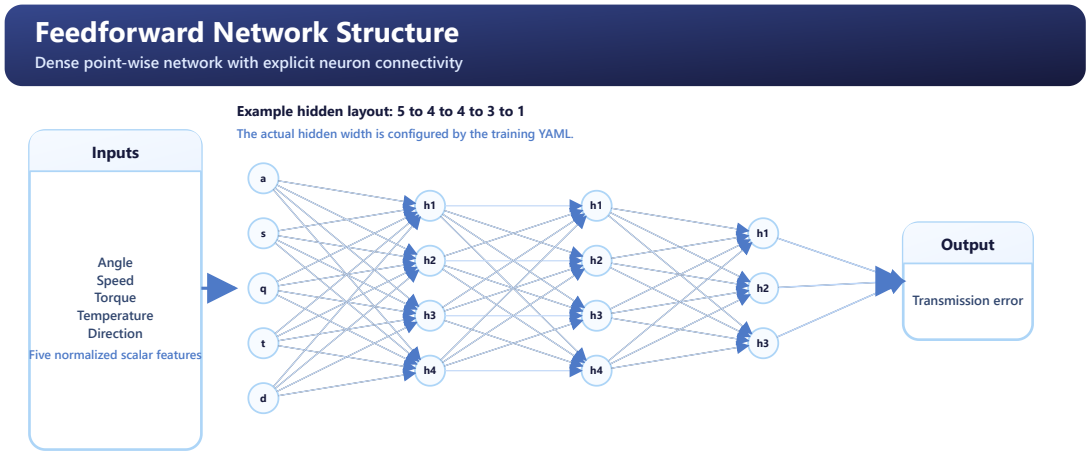
The model can be read as

```
text Input Point -> [angle, speed, torque, temperature, direction] -> normalization -> Linear -> optional LayerNorm -> activation -> optional Dropout -> ... -> Linear -> scalar TE prediction
```

Repository-level schematic

```
text Curve Dataset -> Point Extraction -> TransmissionErrorDataModule -> FeedForwardNetwork -> TransmissionErrorRegressionModule -> Lightning Trainer -> checkpoints + metrics + registries
```

Architecture Diagram



Feedforward architecture diagram

This second diagram shows the network more explicitly as a dense layer stack.

Read it as an example architecture view:

- 5 input features;
- multiple dense hidden layers;
- one scalar output node;
- dense connectivity between consecutive layers.

  The exact hidden widths are YAML-configurable, but the architectural logic remains the same.

Why This Model Exists

This model is the baseline of reference because it is:

- simple;
- flexible;
- easy to train;
- easy to compare against future structured models;
- useful to measure how much gain comes from adding stronger inductive bias later.

  In other words, it answers the question: `How far can we go with a generic static neural regressor before adding TE-specific structure?`

Advantages

- Very simple and robust baseline.
- Can model generic nonlinear couplings among operating variables.
- Low conceptual overhead compared with more structured models.
- Good first benchmark for later temporal, harmonic, or PINN models.
- Easy to deploy or compress compared with heavier architectures.

Disadvantages

- No explicit periodic prior on angular position.
- No explicit decomposition between structured TE component and residual component.
- Less interpretable than harmonic or hybrid structured models.
- May need more data to learn periodic structure that could otherwise be encoded directly.
- Can fit the data well while still giving little physical insight.

Expected Behavior In The TE Context

In the TE problem, this model is expected to work reasonably well when:

- the mapping is smooth enough;
- the available data covers the operating conditions sufficiently;
- the main goal is predictive benchmarking rather than interpretability.

  It is less attractive when the objective is:
- explicit periodic understanding;
- coefficient-level interpretability;
- decomposition into analytical and residual parts;

- physics-oriented reasoning.

Python Model Implementation

The model implementation lives in `scripts/models/feedforward_network.py`.

`get_activation_module(...)`

This helper resolves the configured activation name into a concrete PyTorch activation module.

Its role is simple but important

- keep activation selection declarative in YAML;
- centralize the supported activation set;
- fail early if an unsupported name is requested.

`FeedForwardNetwork.__init__(...)`

This constructor builds the MLP architecture stage by stage.

Main responsibilities

- validate `input_size`, `hidden_size`, `output_size`, and dropout;
- store the architectural parameters for traceability;
- build the hidden stack as repeated:
  - `Linear`
  - optional `LayerNorm`
  - activation
  - optional `Dropout`
- append the final linear output layer.

  The implementation is intentionally explicit rather than abstract. Each hidden stage is assembled in a readable sequence.

`FeedForwardNetwork.forward(...)`

This is the simplest forward path in the current model set.

It just applies

- `self.network(input_tensor)`

  The model therefore expects a normalized point-wise feature tensor and returns a normalized scalar prediction.

Repository Integration

The model is registered in `scripts/models/model_factory.py` under `model_type == "feedforward"`.

The factory path is what makes the model:

- selectable from YAML;
- compatible with the shared training infrastructure;
- compatible with campaign execution and registries.

Training Workflow Overview

The feedforward baseline uses the shared structured-neural training entry point in `scripts/training/train_feedforward_network.py`.

At high level, training works as follows:

1. load YAML configuration;
2. resolve the output artifact identity;
3. build the datamodule and compute normalization statistics;
4. instantiate the model through the factory;
5. wrap it inside the generic regression Lightning module;
6. train with early stopping and checkpointing on `val_mae`;
7. reload the best checkpoint;
8. run final validation and held-out test;
9. save metrics, report, and registry entries.

Training Logic In This Repository

The actual training behavior is split across several files.

`train_feedforward_network.py`

This file is the orchestration entry point.

Important functions

- `print_training_configuration_summary(...)`
  Renders the resolved YAML settings in terminal form.
- `print_dataset_summary(...)`
  Shows the real train/validation/test curve counts and batching setup.
- `print_model_summary(...)`
  Reports trainable, frozen, and total parameter counts.
- `resolve_runtime_config(...)`
  Applies runtime decisions such as disabling `benchmark` when deterministic execution is requested.
- `save_training_test_report(...)`
  Builds the run-local Markdown report after training and testing.
- `train_feedforward_network(...)`
  This is the main orchestration function:
  - prepares the output directory;
  - initializes datamodule, backbone, and regression module;
  - configures logger, checkpointing, early stopping, and progress bar;
  - runs `fit`, `validate`, and `test`;
  - reloads the best checkpoint when available;
  - serializes metrics and updates registries.

`transmission_error_regression_module.py`

This file contains the generic Lightning training logic.

For the feedforward baseline, the most relevant parts are:

- `set_normalization_statistics(...)`

  Copies train-split normalization statistics into Lightning buffers.
- `normalize_input_tensor(...)` and `normalize_target_tensor(...)`

  Move the optimization into normalized space.
- `forward(...)`

  Delegates directly to the MLP on normalized inputs.
- `compute_batch_outputs(...)`

  Produces normalized predictions, denormalized predictions, loss, `MAE`, and `RMSE`.
- `compute_loss(...)`

  Logs the metrics for train, validation, and test phases.
- `configure_optimizers(...)`

  Uses `AdamW`.

`shared_training_infrastructure.py`

This file provides the shared repository plumbing:

- config loading;
- artifact naming;
- output directory resolution;
- datamodule creation;
- model instantiation;
- registry updates;
- common metrics snapshot format.

  For the feedforward baseline, this file is what makes the training workflow reusable and comparable with the newer structured models.

Practical Interpretation

The feedforward model should be read as the neutral nonlinear benchmark of the project:

- stronger than a trivial linear model;
- weaker in inductive bias than harmonic or hybrid structured models;
- useful as the reference point for answering whether added structure really helps.

  If a later structured model beats it clearly, that improvement is meaningful because the feedforward baseline already captures generic nonlinear regression capacity without any TE-specific prior.